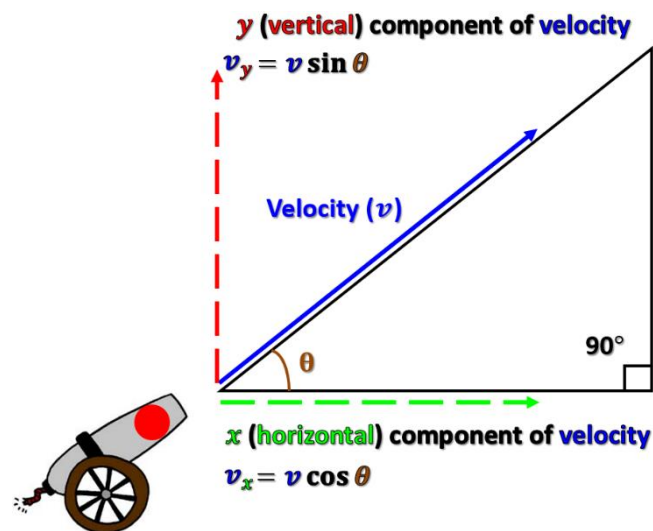


Resolving velocity

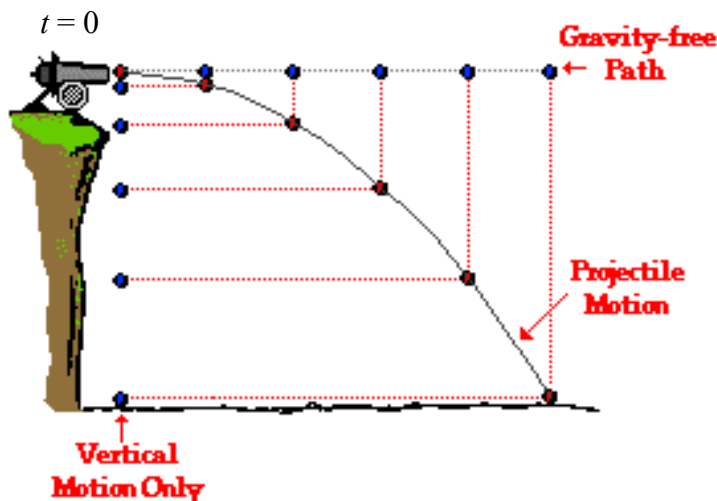
$$v = \sqrt{v_x^2 + v_y^2}$$

$$\frac{v_y}{v_x} = \tan \theta$$



Horizontal projectile motion

- Horizontal and vertical motions are independent of each other.
- Consider velocity, displacement, and acceleration of **vertical** and **horizontal** separately. Only **time** are the **same** in the calculation of both dimensions.



- Horizontal direction:**

$$u_x = v_x = \text{constant}$$

$$a_x = 0$$

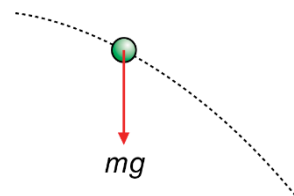
Remarks: There is **no horizontal force** acting on the object, it moves with **uniform velocity** in **horizontal direction**.

- Vertical direction:**

$$u_y = 0 \text{ at } t = 0$$

$$a_y = 9.81 \text{ ms}^{-2}$$

Remarks: The **only force** acting on the object is its **weight**.



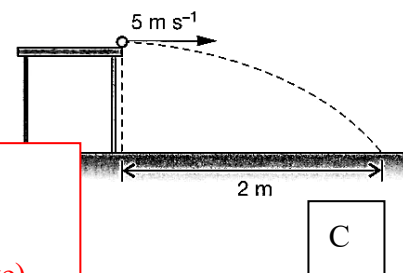
1. A marble rolls off the edge of a horizontal table with a velocity of 5 ms^{-1} . If the marbles hit the floor 2 m away from the spot directly below the edge of the table as shown in the figure, how tall is the table?

- A. 1.05 m
B. 0.95 m
C. 0.78 m
D. 0.7 m

$$t = 2/5 = 0.4 \text{ s}$$

The height of the table is

$$s = ut + \frac{1}{2}at^2 \Rightarrow s_y = \frac{1}{2}(-9.8)(0.4)^2 = -0.784 \text{ m (take upward as +ve)}$$

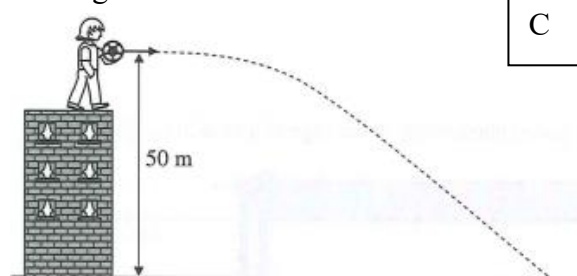


C

2. A child throws a ball horizontally on the top of a building of 50 m tall with an initially velocity 10 ms^{-1} . What is the distance between the ball and the building when the ball hits the ground?

- A. 3.19 m
B. 9.81 m
C. 31.9 m
D. 98.1 m

C
Consider vertical motion,
 $50 = \frac{1}{2}gt^2 \Rightarrow t = \sqrt{\frac{100}{g}} \text{ s}$
Consider horizontal motion,
 $d = vt = 10 \times \sqrt{\frac{100}{g}} = 31.9 \text{ m}$



C

3. In a softball competition, Sharon throws a ball horizontally at a speed of 12 ms^{-1} . The ball touches the ground at a speed of 16 ms^{-1} . What is the time of flight?

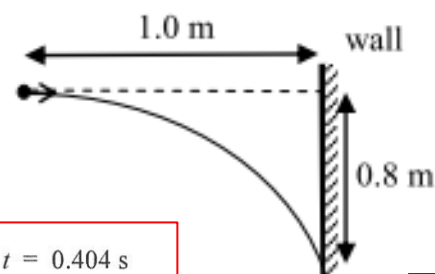
- A. 0.41 s
B. 1.08 s
C. 1.33 s
D. 1.63 s

B
 $v_x = u_x = 12 \text{ m s}^{-1}$
 $v_y^2 = v^2 - v_x^2$
 $= 16^2 - 12^2$
 $v_y = \sqrt{112} \text{ m s}^{-1}$
In the vertical direction, by $v = u + at$,
 $v_y = 0 + gt$
 $t = \frac{v_y}{g} = \frac{\sqrt{112}}{9.81} = 1.08 \text{ s}$

B

4. A particle is projected horizontally towards a vertical wall 1.0 m away. It hits the wall at a position 0.8 m vertically below its point of projection. At what speed is it projected? Neglect air resistance. ($g = 9.81 \text{ ms}^{-2}$)

- A. 2.0 ms^{-1}
B. 2.5 ms^{-1}
C. 5.0 ms^{-1}
D. 6.3 ms^{-1}



Vertical component: $y = \frac{1}{2}gt^2$ $\therefore (0.8) = \frac{1}{2}(9.81)t^2$ $\therefore t = 0.404 \text{ s}$
Horizontal component: $x = ut$ $\therefore (1.0) = u(0.404)$ $\therefore u = 2.5 \text{ m s}^{-1}$

B

5. In a fountain, water is projected horizontally at 3 ms^{-1} from a wall onto a pond which is 1 m below the nozzle. Find the angle to the horizontal at which the projected water enters the pond. Take the upward direction as positive

$$s = ut + \frac{1}{2}at^2$$

$$s_y = \frac{1}{2}(9.81)(t^2) = 1$$

$$t = 0.45 \text{ s}$$

What is the velocity (magnitude and direction) of the water when it enters the pond?

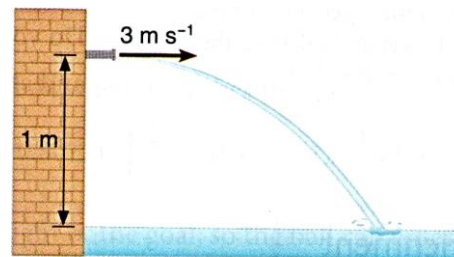
$$v = u + at \Rightarrow v = 0 + (9.81)(0.45) = 4.41 \text{ ms}^{-1}$$

$$v^2 = v_x^2 + v_y^2 = 3^2 + (4.41)^2$$

$$v = 5.33 \text{ ms}^{-1}$$

$$\tan \theta = v_y/v_x = 4.41/3$$

$$\theta = 55.8^\circ \text{ below the horizontal}$$



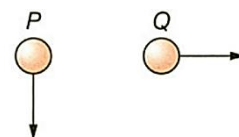
6. Ball P is released from rest while ball Q is projected horizontally from the same height simultaneously. The mass of P is twice the mass of Q .

Which of the following statements is/are correct?

- (1). P and Q reach the ground at the same time.
- (2). P and Q have the same speed on reaching the ground.
- (3). P and Q have the same acceleration throughout their flight in air.

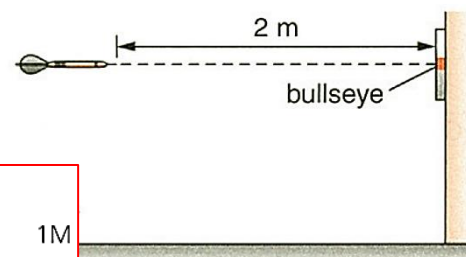
- A. (2) only
- B. (1) and (2) only
- C. (1) and (3) only
- D. All of the above.

C. Q has the same vertical motion as P . Therefore, they reach the ground at the same time.
 $\therefore$ (1) is correct.
 Although the vertical speeds of P and Q are the same, their horizontal speeds are not the same. Therefore their speeds on reaching the ground are not the same.
 $\therefore$ (2) is incorrect.
 P and Q are uniformly accelerated under gravity and the acceleration due to gravity is the same for all objects regardless of their masses.
 $\therefore$ (3) is correct.



7. Mandy throws a dart horizontally towards a dartboard at the same level as the bullseye. The dartboard is 2 m away from her. The dart fails to hit the bullseye and reaches a point 0.1 m below it.

- (a) Find the time of flight of the dart.
- (b) Find the initial velocity of the dart.



(a) In the vertical direction,

$$\text{By } s_y = \frac{1}{2}gt^2,$$

$$t = \sqrt{\frac{2s_y}{g}} = \sqrt{\frac{2 \times 0.1}{9.81}} = 0.143 \text{ s}$$

$\therefore$ The time of flight of the dart is 0.143 s .

(b) In the horizontal direction,

$$\text{initial velocity of the dart} = \frac{s_x}{t}$$

$$= \frac{2}{0.143}$$

$$= 14.0 \text{ m s}^{-1}$$

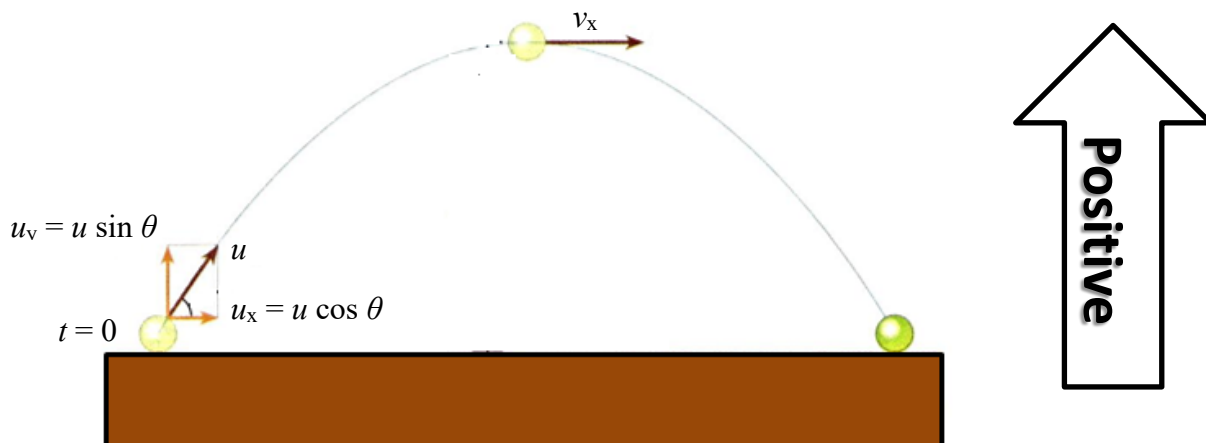
1M

1A

1M

1A

Projectile motion with an angle



● Horizontal direction:

$$u_x = v_x = \text{constant}$$

$$a_x = 0$$

Remarks: There is **no horizontal force** acting on the object, it moves with **uniform velocity** in **horizontal direction**.

● Vertical direction:

$$u_y \neq 0 \text{ at } t = 0$$

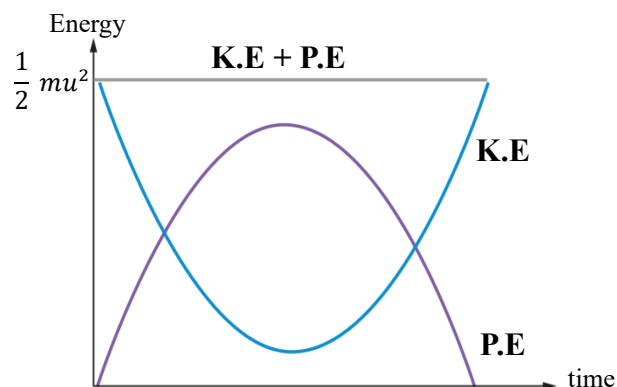
$$a_y = -9.81 \text{ ms}^{-2}$$

$$v_y = 0 \text{ at the highest point}$$

Remarks: The **only force** acting on the object is its **weight**.

Remarks:

- If air resistance is negligible, **total mechanical energy** of projectile is **conserved**.
(K.E + P.E = constant)
- At highest point, K.E $\neq 0$
- The projectile has the **same speed** (not velocity) for both the upward and the downward motion at the **same height**.
- Time of upward flight is equal to downward flight.



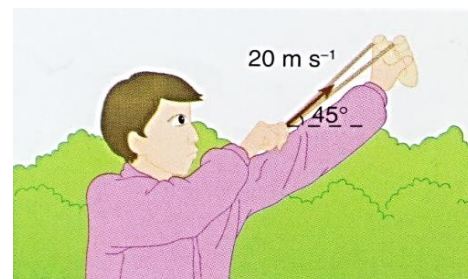
8. A boy uses a catapult to project a stone with a velocity of 20 ms^{-1} at 45° to the horizontal. What are the horizontal and vertical displacements of the stone 0.8 s after the projection? Take the upward direction and the direction to the right as positive.

$$u_x = 14.14 \text{ ms}^{-1}, u_y = 14.14 \text{ ms}^{-1}, t = 0.8 \text{ s}$$

$$s_x = u_x t = 14.14(0.8) = \mathbf{11.3 \text{ m}}$$

$$s = ut + \frac{1}{2}at^2$$

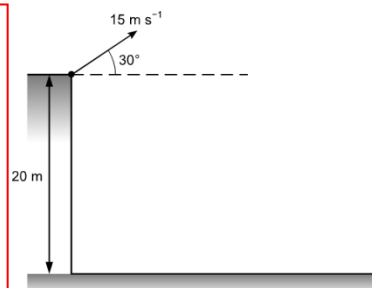
$$s_y = 14.14(0.8) - \frac{1}{2}(9.81)(0.8)^2 = \mathbf{8.17 \text{ m}}$$



9. A ball is kicked off a platform which is 20 m above the ground as shown. Its initial velocity is 15 m s^{-1} and makes an angle of 30° to the horizontal. Eventually the ball lands on the ground. Find the time of flight of the ball.

- A. 1.37 s
B. 2.92 s
C. 3.47 s
D. 5.77 s

$$\begin{aligned} u_x &= v_x = 15 \cos 30^\circ \\ u_y &= 15 \sin 30^\circ, s_y = -20, a_y = -9.81 \\ s &= ut + \frac{1}{2}at^2 \\ -20 &= 15 \sin 30^\circ(t) - \frac{1}{2}(9.81)(t)^2 \\ t &= -1.29 \text{ s (reject) or } 2.92 \text{ s} \end{aligned}$$

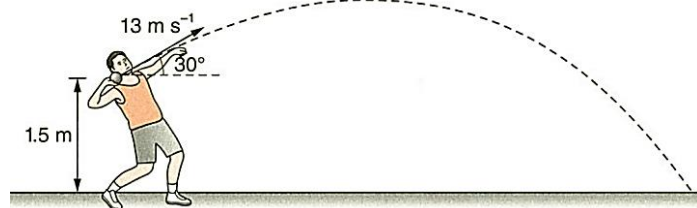


B

10. In a shot-put competition, Charles throws a shot from a height of 1.5 m above the ground at 13 m s^{-1} . The angle of projection is 30° .

- (a) Find the time of flight of the shot.
(b) Charles will win if he can throw the shot 18 m away from him. Will he win with this throw?

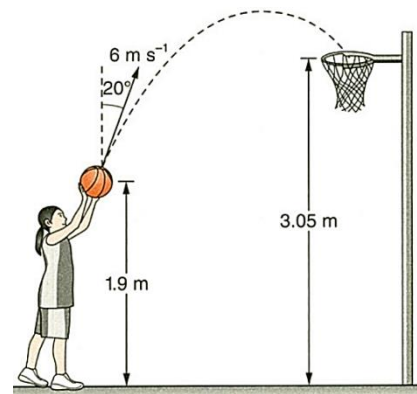
$$\begin{aligned} \text{(a) By } s_y &= u_y t - \frac{1}{2}gt^2, \\ -1.5 &= 13 \times \sin 30^\circ \times t - \frac{1}{2} \times 9.81 \times t^2 \\ 4.905t^2 - 6.5t - 1.5 &= 0 \\ t &= 1.53 \text{ s or } -0.200 \text{ s (rejected)} \\ \therefore \text{ The time of flight of the shot is } &1.53 \text{ s.} \\ \text{(b) In the horizontal direction,} \\ s_x &= u_x t \\ &= 13 \cos 30^\circ \times 1.53 \\ &= 17.2 \text{ m} < 18 \text{ m} \\ \therefore \text{ Charles will not win with this throw.} \end{aligned}$$



11. Daisy shoots a basketball from a height of 1.9 m above the ground with a velocity of 6 m s^{-1} at an angle 20° to the vertical as shown. The ball passes through the hoop from above. The hoop is 3.05 m high.

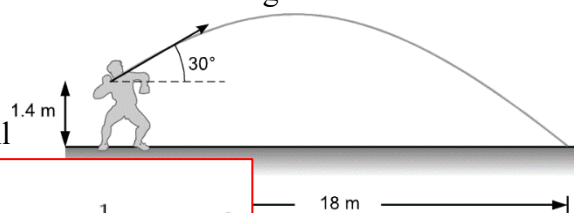
- (a) What is the time needed for the ball to reach the hoop from Daisy?
(b) Calculate the horizontal distance travelled by the ball.

$$\begin{aligned} \text{(a) By } s_y &= u_y t - \frac{1}{2}gt^2, \\ 3.05 - 1.9 &= 6 \cos 20^\circ \times t - \frac{1}{2} \times 9.81 \times t^2 \\ 4.905t^2 - 5.638t + 1.15 &= 0 \\ t &= 0.884 \text{ s or } 0.265 \text{ s (rejected)} \\ \therefore \text{ The time needed for the ball to reach the hoop is } &0.884 \text{ s.} \\ \text{(b) The horizontal distance} \\ &= u_x t \\ &= 6 \sin 20^\circ \times 0.884 \\ &= 1.81 \text{ m} \end{aligned}$$



12. An athlete makes a record of 18 m in a shot put. The ball leaves his hand from a height of 1.4 m above the ground and the angle of projection is 30° .

- (a) What is the time of flight of the ball?
(b) Hence, or otherwise, find the projecting velocity of the ball



Answer: ∴

Let v be the projecting velocity of the ball and t be the time of flight.

- (a) Consider the horizontal displacement. ∴

$$vt \cos 30^\circ = 18$$

$$v = \frac{18}{t \cos 30^\circ} \quad (1)$$

Consider the vertical displacement. ∴

$$-1.4 = vt \sin 30^\circ + \frac{1}{2} \times (-10) \times t^2 \quad (2)$$

Substitute (1) into (2). ∴

$$\begin{aligned} -1.4 &= \left(\frac{18}{t \cos 30^\circ} \right) t \sin 30^\circ + \frac{1}{2} \times (-10) \times t^2 \\ -1.4 &= 18 \tan 30^\circ - 5t^2 \\ t &= 1.5357 \text{ s} \end{aligned}$$

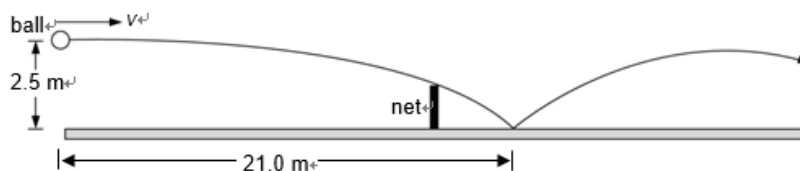
The time of flight is 1.54 s.

- (b) The projecting velocity is ∴

$$\begin{aligned} v &= \frac{18}{1.5357 \times \cos 30^\circ} \\ &\approx 13.5 \text{ m s}^{-1} \end{aligned}$$

13. A tennis machine serves tennis balls over the net for a player to practice return shots. The ball emerges horizontally with velocity v from a height of 2.5 m above the ground. The ball then hits the ground at a point 21.0 m away from the machine as shown in the following diagram. Neglecting air resistance and take the acceleration due to gravity as 9.81 m s^{-2} .

- (a) Show that the ball takes 0.714 s to reach the ground.
(b) Find the velocity v with which the ball left the machine.
(c) Find the vertical component of the velocity of the ball when it reaches the ground.
(d) State the horizontal component of the velocity of the ball when it reaches the ground.
(e) Find the velocity at which the ball strikes the ground.
(f) What would be the effect on the direction of the velocity in (e) if air resistance is taken into account?



Take the downward direction as positive.

- (a) In the vertical direction:

$$\text{By } s_y = u_y t + \frac{1}{2} g t^2,$$

$$2.5 = 0 + \frac{1}{2} (9.81) t^2$$

$$t = 0.714 \text{ s}$$

$$(d) \quad 29.4 \text{ m s}^{-1}.$$

- (b) In the horizontal direction:

$$v = \frac{s_x}{t}$$

$$= \frac{21.0}{0.714}$$

$$= 29.4 \text{ m s}^{-1}$$

$$(e) \quad v = \sqrt{v_x^2 + v_y^2} = \sqrt{7^2 + 29.4^2} = 30.2 \text{ m s}^{-1}$$

$$\tan \theta = \frac{v_y}{v_x} = \frac{7}{29.4}$$

$$\theta = 13.4^\circ$$

∴ The velocity of the ball is 30.2 m s^{-1} at 13.4° below the horizontal.

- (c) In the vertical direction:

$$\text{By } v_y = u_y + gt,$$

$$v_y = 0 + 9.81 \times 0.714$$

$$= 7.00 \text{ m s}^{-1}$$

∴ The vertical component is 7.00 m s^{-1} .

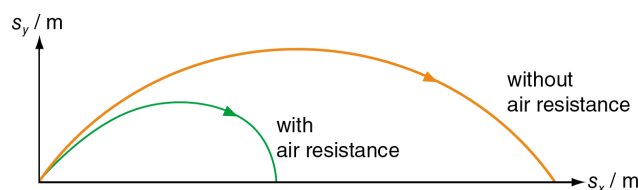
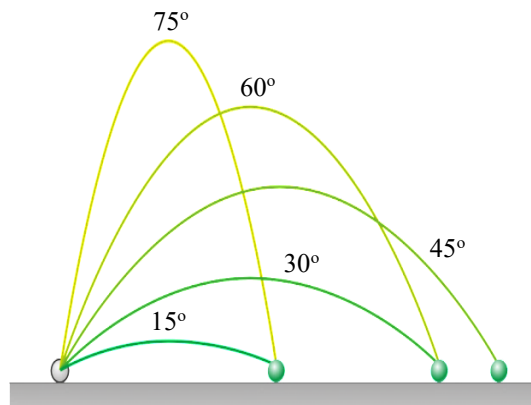
- (f) The velocity will make a larger angle with the horizontal.

Horizontal Range R :

$$R = \frac{u^2 \sin 2\theta}{g}$$

Remarks:

- R is maximum when $\theta = 45^\circ$
- Two projection angles give the same range
($\theta, 90^\circ - \theta$)
- In the **presence of air resistance**, a projectile has:
 - (1) An asymmetric trajectory
 - (2) A reduced maximum height and range.



14. A ball is projected at a velocity of 20 m s^{-1} which makes an angle of 30° to the horizontal. What is the range of the ball?

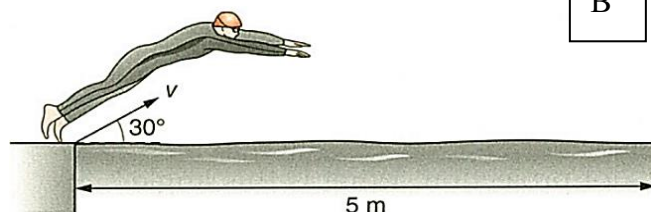
- A. $\frac{20 \sin 30^\circ}{g}$
 B. $\frac{400 \sin 60^\circ}{g}$
 C. $\frac{400 \sin^2 30^\circ}{g}$
 D. $\frac{400 \sin^2 60^\circ}{g}$

B

15. Terry jumps from the pool side into the swimming pool at a velocity v at an angle of 30° to the horizontal. He enters the water at a position 5 m away from the pool side. What is the velocity v ?

- A. 7.00 m s^{-1}
 B. 7.53 m s^{-1}
 C. 8.09 m s^{-1}
 D. 9.90 m s^{-1}

$$\begin{aligned} \text{Range} &= \frac{u^2 \sin 2\theta}{g} \\ 5 &= \frac{v^2 \sin 2(30^\circ)}{9.81} \\ v &= 7.53 \text{ m s}^{-1} \end{aligned}$$



B

16. A football is kicked away at an angle 42° to the horizontal with an initial velocity of 40 m s^{-1} . Find the range of the football when it returns to the ground.

- A. 177 m
 B. 162 m
 C. 88.5 m
 D. 35.8 m

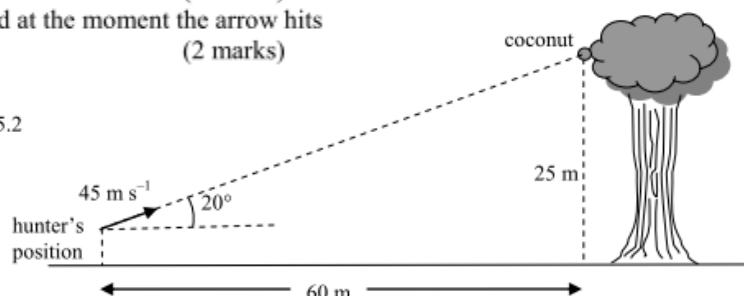
$$\begin{aligned} &\text{Take upward as positive, then,} \\ &u_y = 40 \sin 42^\circ \text{ m s}^{-1}, v_y = -40 \sin 42^\circ \text{ m s}^{-1}, \\ &g = -10 \text{ m s}^{-2} \\ &v_y = u_y + gt \\ &-40 \sin 42^\circ = 40 \sin 42^\circ + (-10) \times t \\ &t = 5.353 \text{ s} \\ &\therefore \text{ The range of the football is:} \\ &R = u_x t = 40 \cos 42^\circ \times 5.353 = 159 \text{ m} \end{aligned}$$

B

17. *(b) The hunter stands at about 60 m away from a tree as shown in Figure 5.2. He uses the bow to release the arrow in order to shoot a coconut held by a monkey (not shown in the figure) in the tree. The coconut is at a height of 25 m from the ground. The hunter aims directly at the coconut and the arrow leaves the bow at a speed of 45 m s^{-1} making an angle of 20° to the horizontal. At the moment the hunter releases the arrow, the monkey drops the coconut such that it falls vertically from rest. Neglect air resistance and the arrow's size. ($g = 9.81 \text{ m s}^{-2}$)

- (i) Find the time taken for the arrow to hit the coconut. (2 marks)
(ii) Find the height of the coconut from the ground at the moment the arrow hits it. (2 marks)

Figure 5.2

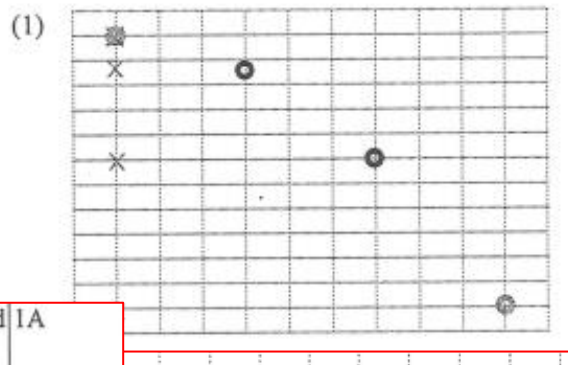


(b) (i)	$d = v \cos 20^\circ t$ $60 = 45 \cos 20^\circ t$ $t = 1.42 \text{ s}$ (Accept 1.40 s to 1.42 s)	1M correct eqn. or with sub. 1A correct answer	1M 1A
(ii)	$h = 25 - \frac{1}{2}gt^2$ $= 25 - \frac{1}{2}(9.81)(1.42)^2$ $= 15.1 \text{ m}$ [or $h = 14.9 \text{ m}$] (Accept 14.9 m to 15.4 m)	1M correct eqn. ($s = \frac{1}{2}gt^2$) to find the falling distance of coconut initial velocity must be zero 1A correct answer [$h = 14.9 \text{ m}$ for $g = 10 \text{ m s}^{-2}$]	1M 1A

18. The bearing is projected horizontally instead of released from rest. The bearing is projected at time $t = 0$, and a stroboscopic photo is taken at 0.05 s time intervals. The first and the last image of the stroboscopic photo are shown using circles (O) in Figure 4.2. For reference, the stroboscopic photo of the bearing released from rest is also shown in the figure using cross (X).

- (a)(1) In figure 4.2, mark the positions of the projected bearing in the stroboscopic photo using circles (O).
(2) Given that the bearing is projected horizontally with an initial speed of 1 ms^{-1} , using the results of (a)(i) to calculate the speed of the projected bearing when the last image was taken.
(b) If a small ball is released from rest from the top of a cliff, the speed of the ball becomes constant after a period of time. By considering the forces acting on the ball and using Newton's law of motion, explain why the speed of the ball becomes constant.

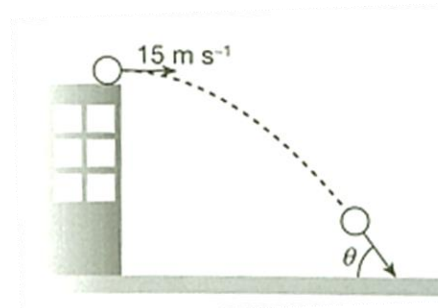
(2) $v_x = 1 \text{ m s}^{-1}$
 $v_y = u_y + gt$
 $= 0 + 9.78 \times (0.05 \times 3)$
 $= 1.47 \text{ m s}^{-1}$
 $v = \sqrt{v_x^2 + v_y^2}$
 $= \sqrt{1^2 + 1.47^2}$
 $= 1.78 \text{ m s}^{-1}$



- (b) The air resistance acting on the ball increases as its speed increases. 1A
When the air resistance equals to the weight of the ball, 1A
net force acting on the ball becomes zero, by Newton's first law of motion, the ball travels with constant speed. 1A
Or
net force acting on the ball becomes zero, by Newton's second law of motion, the ball will not accelerate further and travels with constant speed. 1A

19. In the figure, a ball is projected with a horizontal velocity of 15 m s^{-1} from the top of a building at time $t = 0$. It hits the ground at an angle θ with the horizontal at $t = 4 \text{ s}$. Neglect air resistance.

- At $t = 1 \text{ s}$, find the horizontal speed and vertical speed of the ball.
- Find the horizontal displacement and vertical displacement of the ball just before hitting the ground.
- At $t = 4 \text{ s}$, find the velocity of the ball.
- Find the size of θ



Take the downward direction as positive.

(a) Horizontal motion: $u_x = 15 \text{ m s}^{-1}$ (constant)

Vertical motion: $u_y = 0 \text{ m s}^{-1}$

$$a_y = 9.81 \text{ m s}^{-2}$$

Apply the equation $v_y = u_y + a_y t$,

$$\begin{aligned} v_y &= 0 + (9.81)(1) \\ &= 9.81 \text{ m s}^{-1} \end{aligned}$$

(b) Horizontal motion: $u_x = 15 \text{ m s}^{-1}$ (constant)

Apply the equation $s_x = u_x t$,

$$\begin{aligned} s_x &= (15)(4) \\ &= 60 \text{ m} \end{aligned}$$

Vertical motion: $u_y = 0 \text{ m s}^{-1}$

$$a_y = 9.81 \text{ m s}^{-2}$$

Apply the equation $s_y = u_y t + \frac{1}{2} a_y t^2$,

$$\begin{aligned} s_y &= 0 + \frac{1}{2} (9.81)(4)^2 \\ &= 78.48 \text{ m} \end{aligned}$$

(c) Horizontal motion: $u_x = 15 \text{ m s}^{-1}$ (constant)

Vertical motion: $u_y = 0 \text{ m s}^{-1}$

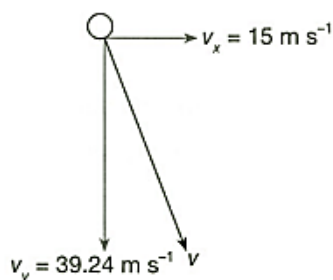
$$a_y = 9.81 \text{ m s}^{-2}$$

Apply the equation $v_y = u_y + a_y t$,

$$\begin{aligned} &= 0 + (9.81)(4) \\ &= 39.24 \text{ m s}^{-1} \end{aligned}$$

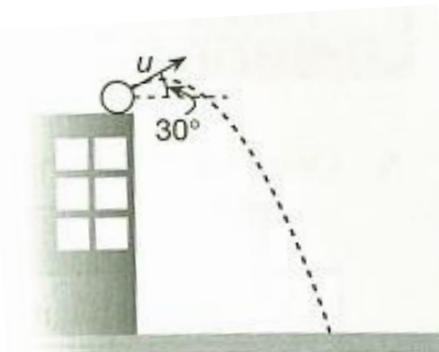
$$\begin{aligned} \text{Hence the velocity } v &= \sqrt{15^2 + 39.24^2} \\ &= 42.0 \text{ m s}^{-1} \end{aligned}$$

$$\begin{aligned} \text{(d) } \tan \theta &= \frac{39.24}{15} \\ \theta &= 69.1^\circ \end{aligned}$$



20. In the figure, a stone is projected with a speed u that makes an angle of 30° with the horizontal from the top of a tower of height 100 m. It strikes the ground 120 m away from the foot of the tower.

- (a) Find the time taken for the stone to reach the ground.
(b) Find u .
(c) What is the speed of the stone just before hitting the ground?



Take the downward direction as positive.

- (a) Horizontal motion: $u_x = u \cos 30^\circ$ (constant)

$$s_x = u_x t$$

$$120 = (u \cos 30^\circ) t$$

$$u = \frac{120}{t \cos 30^\circ} \dots\dots\dots (1)$$

- Vertical motion: $u_y = -u \sin 30^\circ$

$$a_y = 9.81 \text{ m s}^{-2}$$

$$s_y = 100 \text{ m}$$

$$s_y = u_y t + \frac{1}{2} a t^2$$

$$100 = (-u \sin 30^\circ) t + \frac{1}{2} (9.81) t^2 \dots\dots\dots (2)$$

Substitute (1) into (2),

$$100 = -120 \tan 30^\circ + \frac{1}{2} (9.81) t^2$$

$$t = 5.87 \text{ s}$$

- (b) Substitute the result of (a) into (1),

$$u = \frac{120}{t \cos 30^\circ}$$

$$= 23.6 \text{ m s}^{-1}$$

- (c) By the law of conservation of energy,

$$\frac{1}{2} m v^2 = \frac{1}{2} m u^2 + m g h$$

$$v^2 = u^2 + 2 g h$$

$$v = 50.2 \text{ m s}^{-1}$$

